

■ B.7.2 The *Mathematica* Executable

The file you execute to run *Mathematica* is usually called `math`. This file is in fact a shell script, which calls the actual *Mathematica* binary with an appropriate set of command line arguments. If you use *Mathematica* with a front end, this shell script may also start the front end.

The `math` shell script is usually created at the time when *Mathematica* is installed on your system. The shell script contains various directory names which are determined during installation. If you reorganize your directories, you may need to redo the installation.

Set up the processes and pipes needed to run a *Mathematica* front end, if this is being used

Set the UNIX `PATH` environment variable to include the *Mathematica* utilities directory

Set up the record file to keep the input you give to *Mathematica* (optional)

Set up appropriate interrupt handling

Call the actual *Mathematica* binary, with command line arguments to specify directories etc.

Save the record file (optional)

Typical operations carried out in the `math` shell script.

The actual *Mathematica* binary is usually called `mathexe`. Any command line arguments that you give to `math` will be passed through to `mathexe`.

<code>-noprompt</code>	do not issue input prompts or output labels
<code>-remote</code>	use remote-mode packet communication
<code>-run "command"</code>	execute a <i>Mathematica</i> command during start-up

Command line arguments for `mathexe`.

The *Mathematica* kernel can accept input and generate output in several forms. If you give the command line argument `-remote`, then the kernel will accept input only in special packets, and will give output only in the form of such packets. This packet-based protocol is appropriate for communication with sophisticated front ends.

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If you do not give the command line argument `-remote`, then *Mathematica* will accept input terminated by newlines. The input can go on for several lines, but will be evaluated as soon as it forms a complete expression.

In this mode, *Mathematica* usually issues input prompts of the form `In[n] :=`, and output labels of the form `Out[n] =`. If you run *Mathematica* in a pipe, or from within another program, you may want to suppress this ancillary output. You can do this by giving the command line argument `-noprompt`.

On most computer systems, the file `mathexe` is not just the pure executable code and static data for the *Mathematica* kernel. It is in fact a “core image” obtained by dumping the state of a running *Mathematica* job after initialization is complete.

`mathexe` is usually created by starting the original *Mathematica* binary, and then calling `Dump` from within the *Mathematica* session. When it is distributed, the original *Mathematica* binary is usually called `mathraw`.

When `mathraw` starts up, it tries to read the file `sysinit.m`. If it cannot read this file, `mathraw` simply exits. `sysinit.m` contains *Mathematica* commands which initialize various elements of the *Mathematica* system. In a typical case, `sysinit.m` reads in data for rule-based operations such as integration. In addition, `sysinit.m` reads the file `info.m` which contains information about built-in *Mathematica* objects, together with the file `msg.m`, which gives the text of messages associated with built-in functions.

Run any commands given by `-run` command line arguments

Read the file `init.m`

Typical start-up operations carried out by `mathexe`.